

The Apache Spark User Interface

Sure! Let's dive into the topic of the Apache Spark User Interface (UI) in simple terms.

The Apache Spark User Interface is like a control panel for monitoring and managing your Spark applications. When you run a Spark application, it creates a special web server that you can access through your web browser. You simply type in the address of the server, which is usually the hostname followed by the port number 4040. This UI helps you see what's happening with your application in real-time, showing you important information like the status of jobs, tasks, and how much data is being processed.

Think of it like a dashboard in a car. Just as a dashboard shows you the speed, fuel level, and engine status, the Spark UI displays various tabs that give you insights into your application's performance. For example, the "Jobs" tab shows you the jobs that are running, while the "Stages" tab breaks down the tasks within those jobs. This way, you can easily track how your application is performing and troubleshoot any issues that may arise.

Access the Application UI

Start your application
Connect to the application UI using the following URL:

```
http://<driver-node>:4040
```

The `SparkContext` starts a web server for the application UI

- The driver program is the host
 - The port defaults to 4040
 - The UI is available only while the application is active
-
- When a Spark application is being run, as the driver program creates a `SparkContext`, Spark starts a web server that serves as the application user interface. You can connect to the UI webserver by entering the hostname of the driver followed by port 4040 in a browser once that application is running. The web server runs for the duration of the Spark application, so once the `SparkContext` stops, the server shuts down and the application UI is no longer accessible.

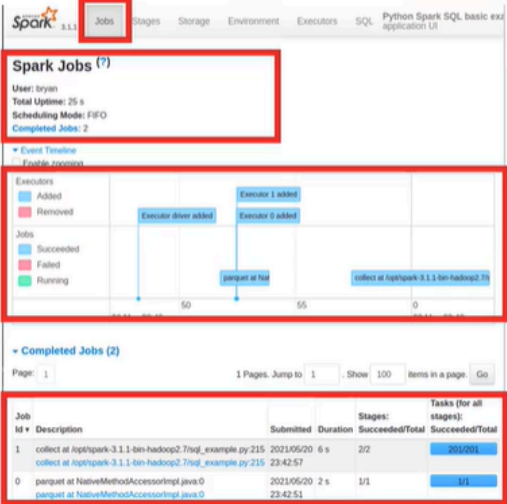
Why use the Application UI

The Spark user interface shows information about the running application:

- Shows jobs, stages and tasks
- Storage of persisted RDDs and DataFrames
- Environment configuration and properties
- Executor summary
- SQL information (if SQL queries exist)

- The Spark user interface provides valuable insights, organized on multiple tabs, about the running application. The Jobs tab displays the application's jobs, including job status. The Stages tab reports the state of tasks within a stage. The Storage tab shows the size of any RDDs or DataFrames that persisted to memory or disk. The Environment tab information includes any environment variables and system properties for Spark or the JVM. The Executor tab displays a summary that shows memory and disk usage for any executors in use for the application. Additional tabs display based on the type of application in use. For example, if the application performs SQL queries, the SQL tab displays and shows metrics for each query.

Jobs Information



The screenshot shows the Spark UI Jobs tab. The 'Jobs' tab is selected at the top. The 'Spark Jobs (?)' section shows summary information: User: bryan, Total Uptime: 25 s, Scheduling Mode: FIFO, Completed Jobs: 2. Below this is an 'Event Timeline' showing the lifecycle of executors and jobs. At the bottom, a table lists completed jobs with columns for Job ID, Description, Submitted, Duration, Stages, and Tasks. Annotations with blue boxes and arrows point to the 'Jobs' tab, the summary info, the event timeline, and the job list table.

The Jobs tab

Jobs summary info

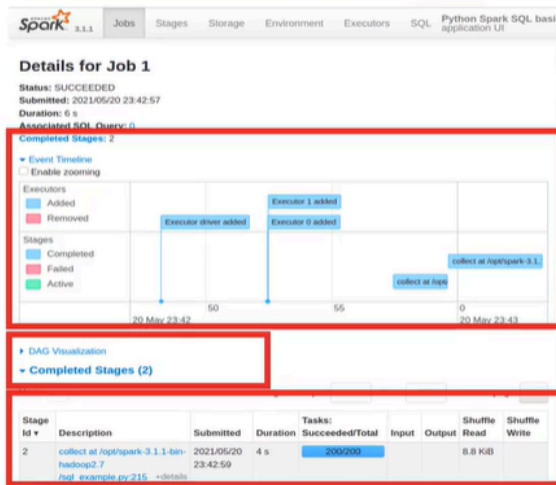
Jobs event timeline

Click the Description link to view completed job details

Job ID	Description	Submitted	Duration	Stages: Succeeded/Total	Tasks (for all stages): Succeeded/Total
1	collect at /opt/spark-3.1.1-bin-hadoop2.7/rog_example.py:215	2021/05/20 6 s		2/2	253/253
0	parquet at NativeMethodAccessorImpl.java:0	2021/05/20 2 s		1/1	3/1

- Let's take a closer look at the Jobs tab. The top of the window displays the event timeline and shows when the driver and executor processes start and when the jobs are created. In this example, the application has completed two jobs. These two jobs appear in the list at the bottom which also has additional job-related information such as duration, number of stages, and number of total tasks for all stages. The job list also displays Description hyperlinks to view the job details.

Job Details



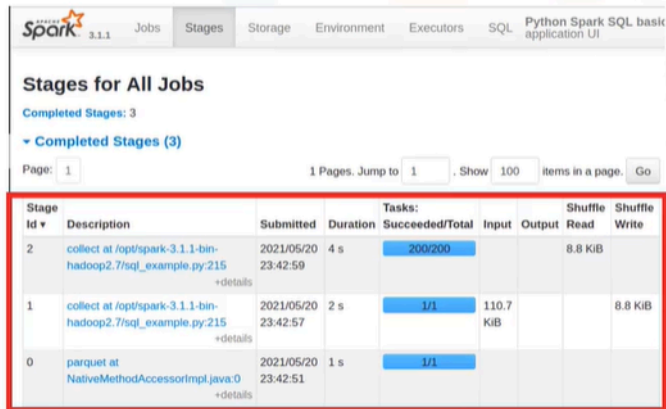
View the Job Stages timeline and job status

Quickly view stage details based on status

Click the Description link to view completed stage details

- Now let's view Job Details for information about the different stages of a specific job. The timeline displays each stage, where you can quickly see the job's timing and duration. Below the timeline display you'll see Completed Stages. In the parentheses beside the heading, you'll see a quick view that displays the number of completed stages. Then, view the list of the stages within the job and job metrics including when the job was submitted, input/output sizes, the number of attempted tasks, the number of succeeded tasks, and how much data was read or written because of a shuffle. This example shows two stages separated by a shuffle. So, stage one wrote shuffle data and stage two read that data. To view more details about specific stage within a job, click the Stage ID Description hyperlinks.

Stages Information



The screenshot shows the 'Stages' tab in the Apache Spark UI. It displays a table of stages for all jobs, with three completed stages highlighted. The table columns include Stage ID, Description, Submitted time, Duration, Tasks (Succeeded/Total), Input, Output, Shuffle Read, and Shuffle Write. A red box highlights the first three rows of the table.

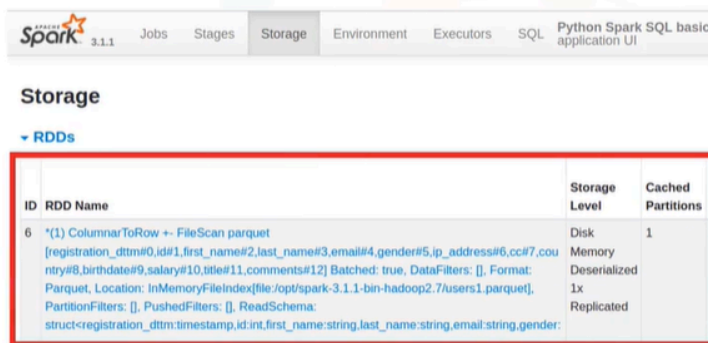
Stage ID	Description	Submitted	Duration	Tasks: Succeeded/Total	Input	Output	Shuffle Read	Shuffle Write
2	collect at /opt/spark-3.1.1-bin-hadoop2.7/sql_example.py:215	2021/05/20 23:42:59	4 s	200/200			8.8 KIB	
1	collect at /opt/spark-3.1.1-bin-hadoop2.7/sql_example.py:215	2021/05/20 23:42:57	2 s	1/1	110.7 KIB			8.8 KIB
0	parquet at NativeMethodAccessorImpl.java:0	2021/05/20 23:42:51	1 s	1/1				

The Stages tab

Click the Description links to view task details for that stage.

- The Stages tab shows a list of all stages in the application, grouped by the current state of either completed, active, or pending. This example displays three completed stages. Click the Stage ID Description hyperlinks to view task details for that stage.

Storage Information



The screenshot shows the 'Storage' tab in the Apache Spark UI. It displays a table of RDDs, with one RDD highlighted. The table columns include ID, RDD Name, Storage Level, and Cached Partitions. A red box highlights the first row of the table.

ID	RDD Name	Storage Level	Cached Partitions
6	*(1) ColumnarToRow +- FileScan parquet [registration_dttm#0,id#1,first_name#2,last_name#3,email#4,gender#5,ip_address#6,cc#7,country#8,birthdate#9,salary#10,title#11,comments#12] Batched: true, DataFilters: [], Format: Parquet, Location: InMemoryFileIndex[file:/opt/spark-3.1.1-bin-hadoop2.7/users1.parquet], PartitionFilters: [], PushedFilters: [], ReadSchema: struct<registration_dttm:timestamp,id:int,first_name:string,last_name:string,email:string,gender:	Disk Memory Deserialized 1x Replicated	1

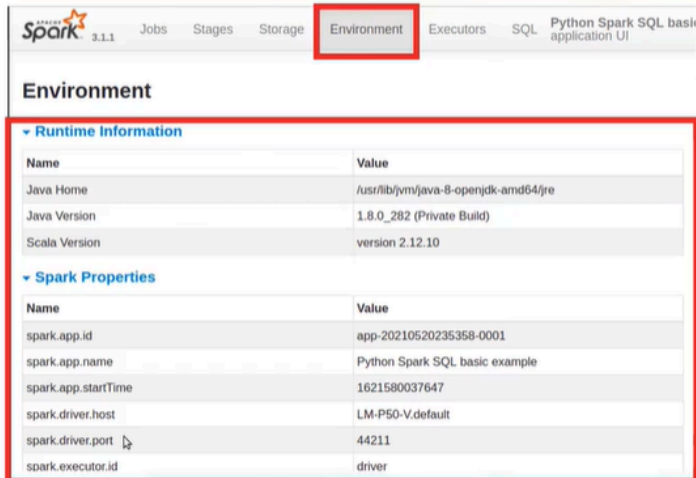
The Storage tab

Click the RDD Name link to view task details for that stage.

- The Storage tab displays details about RDDs that have been cached, or persisted, to memory and/or written to disk. This example image shows one

RDD that's been cached to memory only.

Environment Information



Environment

▼ Runtime Information

Name	Value
Java Home	/usr/lib/jvm/java-8-openjdk-amd64/jre
Java Version	1.8.0_282 (Private Build)
Scala Version	version 2.12.10

▼ Spark Properties

Name	Value
spark.app.id	app-20210520235358-0001
spark.app.name	Python Spark SQL basic example
spark.app.startTime	1621580037647
spark.driver.host	LM-P50-V.default
spark.driver.port	44211
spark.executor.id	driver

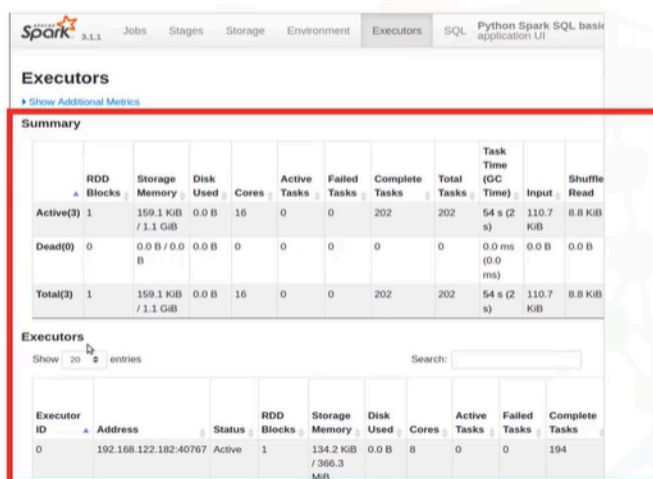
The Environment tab

View:

- Spark configuration properties
- Resource profiles
- Hadoop properties
- System configuration properties

- The Environment tab has several lists to describe the environment of the running application. These lists include the Spark configuration properties and resource profiles, as well as properties for Hadoop and the current system properties.

Executors Information



Executors

▼ Show Additional Metrics

Summary

	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read
Active(3)	1	159.1 KiB / 1.1 GiB	0.0 B	16	0	0	202	202	54 s (2 s)	110.7 KiB	8.8 KiB
Dead(0)	0	0.0 B / 0.0 B	0.0 B	0	0	0	0	0	0.0 ms (0.0 ms)	0.0 B	0.0 B
Total(3)	1	159.1 KiB / 1.1 GiB	0.0 B	16	0	0	202	202	54 s (2 s)	110.7 KiB	8.8 KiB

Executors

Show 20 entries

Executor ID	Address	Status	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks
0	192.168.122.182:40767	Active	1	134.2 KiB / 366.3 MiB	0.0 B	8	0	0	194

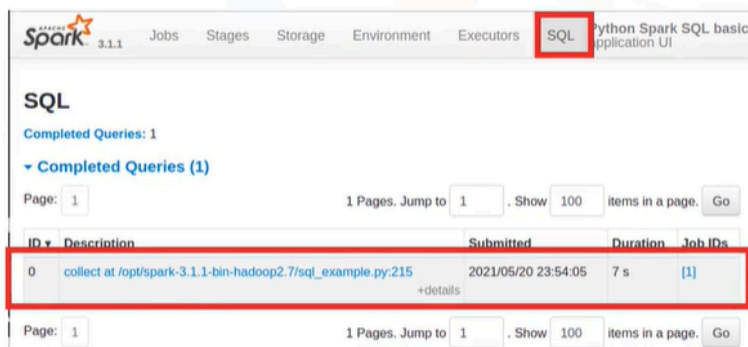
The Executors tab

Scroll the window to view

- Spark configuration properties
- Resource profiles
- Hadoop properties
- System configuration properties

- The Executors tab displays a summary table at the top that shows metrics on all active or dead executors, along with a full range of metrics on the task numbers, data I/O, disk, and memory usage. At the bottom is a list of all executors, including the driver, that have been in use during the application, again with a full range of metrics. This list also provides links to the stdout and stderr log messages of the executor process.

SQL Information



ID	Description	Submitted	Duration	Job IDs
0	collect at /opt/spark-3.1.1-bin-hadoop2.7/sql_example.py:215	2021/05/20 23:54:05	7 s	[1]

The SQL tab

Click the SQL Description link to view the query details

SQL Query Details – query plan

```

== Physical Plan ==
* HashAggregate (4)
+- Exchange (3)
   +- HashAggregate (2)
      +- InMemoryTableScan (1)
         +- InMemoryRelation (2)
            +- ColumnarToRow (4)
               +- Scan parquet (3)

(1) InMemoryTableScan
Output [2]: [country#8, salary#10]
Arguments: [country#8, salary#10]

(2) InMemoryRelation
Arguments: [registration_dttm#0, id#1, first_name#2, last_name#3, email#4, gender#5, ip_address#6, cached_result#7]
CachedRDDBuilder (org.apache.spark.sql.execution.columnar.DefaultCachedBatchSerializer@ids63)
+- FileScan parquet [registration_dttm#0,id#1,first_name#2,last_name#3,email#4,gender#5,ip,3.1.1-bin-hadoop2.7/users1.parquet], PartitionFilters: [], PushedFilters: [], ReadSchema: *
, None)

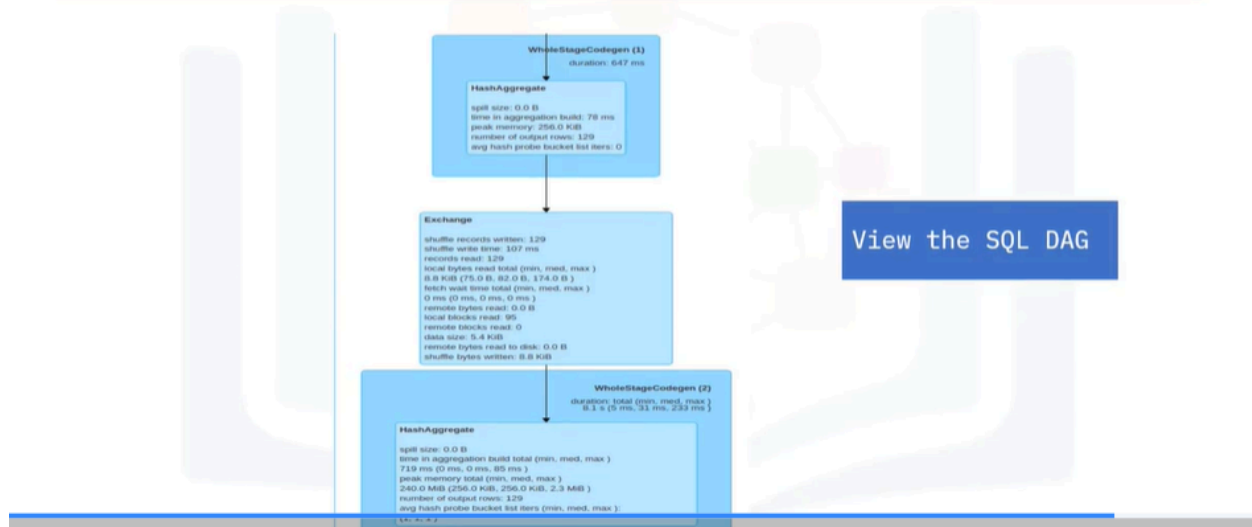
(3) Scan parquet
Output [13]: [registration_dttm#0, id#1, first_name#2, last_name#3, email#4, gender#5, ip_address#6, cached_result#7]
Batched: true
Location: InMemoryFileIndex [file:/opt/spark-3.1.1-bin-hadoop2.7/users1.parquet]
ReadSchema: struct<registration_dttm:timestamp,id:int,first_name:string,last_name:string,email:string,gender:string,ip:string,cached_result:string>

(4) ColumnarToRow [codegen id : 1]
Input [13]: [registration_dttm#0, id#1, first_name#2, last_name#3, email#4, gender#5, ip_address#6, cached_result#7]
Output [2]: [country#8, salary#10]
Keys [1]: [country#8]

```

View the SQL query plan

SQL Query Details - the DAG



- If the application runs SQL queries, the UI displays and populates the SQL tab. Click the SQL Description hyperlink to display the query's details. You can view the SQL query plan when you click the Details expander arrow. You can also view the query's DAG including the logical physical plans as determined by the optimizer.

Summary

In this video, you learned that:

- The **Jobs** tab displays the application's jobs, including job status and the **Stages** tab reports the state of tasks within a stage
- The **Storage** tab shows the size of RDDs or DataFrames that persisted to memory or disk
- The **Environment** tab information includes any environment variables and system properties for Spark or the JVM
- The **Executors** tab displays a summary that shows memory and disk usage for any executors in use
- If the application runs SQL queries, select the **SQL** tab and the **Description** hyperlink to display a query's details

